LOCALLY TRIVIAL FAMILIES FROM FIBRE EQUIVALENCES

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Kernel base change. lemma-base-change-rek L26476 (carded this batch). Restricts and pulls back relative equivalence kernels.
- Composition compatibility. lemma-base-change-is-functor L26480 (carded this batch). Composition passes through base change.
- Rigidity lemma. lemma-no-deformations L26456 (carded this batch). Concludes triviality over a neighbourhood.

Source reference. Original definition 0G0I: sources/stacks-equiv/pinned-original-source.tex, source lines 3792–3792. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G0G: sources/stacks-equiv/pinned-original-source.tex, source lines 3729–3729. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G0K: sources/stacks-equiv/pinned-original-source.tex, source lines 3838–3838. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G0R: sources/stacks-equiv/pinned-original-source.tex, source lines 4131–4131. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 0G0F: sources/stacks-equiv/pinned-original-source.tex, source lines 3627–3627. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be an algebraically closed field. Let X be a smooth proper scheme over k. Let $f : Y \to S$ be a smooth proper morphism with S of finite type over k. Let K be the Fourier-Mukai kernel of a relative equivalence from $X \times S$*

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to Y over S. Then S can be covered by open subschemes U such that there is a U-isomorphism $f^{-1}(U) \cong Y_0 \times U$ for some Y_0 proper and smooth over k.

Proof. Choose a closed point $s \in S$. Since k is algebraically closed this is a k-rational point. Set $Y_0 = Y_s$. The restriction K_0 of K to $X \times Y_0$ is the Fourier-Mukai kernel of a relative equivalence from X to Y_0 over $\mathrm{Spec}(k)$ by Lemma 0G0K. Let K_0' in $D_{perf}(\mathcal{O}_{Y_0 \times X})$ be the object assumed to exist in Definition 0G0I. Then K_0' is the Fourier-Mukai kernel of a relative equivalence from Y_0 to X over $\mathrm{Spec}(k)$ by the symmetry inherent in Definition 0G0I. Hence by Lemma 0G0K we see that the pullback

$$M = (Y_0 \times X \times S \to Y_0 \times X)^* K_0'$$

on $(Y_0 \times S) \times_S (X \times S) = Y_0 \times X \times S$ is the Fourier-Mukai kernel of a relative equivalence from $Y_0 \times S$ to $X \times S$ over S. Now consider the kernel

$$K_{new} = R\mathrm{pr}_{13,*}(L\mathrm{pr}_{12}^* M \otimes^{\mathbf{L}}_{\mathcal{O}_{(Y_0 \times S) \times_S (X \times S) \times_S Y}} L\mathrm{pr}_{23}^* K)$$

on $(Y_0 \times S) \times_S Y$. This is the Fourier-Mukai kernel of a relative equivalence from $Y_0 \times S$ to Y over S since it is the composition of two invertible arrows in the category constructed in Section 0G0F. Moreover, this composition passes through base change (Lemma 0G0G). Hence we see that the pullback of K_{new} to $((Y_0 \times S) \times_S Y)_s = Y_0 \times Y_0$ is equal to the composition of K_0 and K_0' and hence equal to the identity in this category. In other words, we have

$$L(Y_0 \times Y_0 \to (Y_0 \times S) \times_S Y)^* K_{new} \cong \Delta_{Y_0/k,*} \mathcal{O}_{Y_0}$$

Thus by Lemma 0G0R we conclude that $Y \to S$ is isomorphic to $Y_0 \times S$ in an open neighbourhood of s. This finishes the proof. □